

Liquid Crystals

Liquid crystals are a unique state of matter, between solid (crystalline) and liquid (isotropic) phases some compounds form a distinct, different intermediate phase, sometimes referred to as the “fourth state of matter” or “mesophase”.

These compounds display properties of both solid and liquid.

Certain solids on heating undergo two sharp phase transformations; firstly they fuse to turbid liquid and then at higher temperature converted into a clear liquid. The turbid/ translucent state is called liquid crystals / mesomorphic state.

Solid-----> liquid crystal----->clear liquid state

The liquid crystals show anisotropy i.e. they have different physical properties in different directions. Also they exhibit double refraction and give interference patterns in polarized light. Thus the structures of liquid crystal are in between of liquid and a crystalline solid. They show properties of liquids like mobility, surface tension, viscosity etc. and optical properties of crystalline solids.

e.g. p-azoxy anisole, p-azoxyphenetole, Cholesteryl benzoate etc

The study of Liquid crystal started in year 1888 AD by Austrian scientist Friedrich Reinitzer.

The liquid crystals can be classified on the basis of their structure or molecular arrangement into 3 types:

Smectic liquid crystals.

Nematic liquid crystals &

Cholesteric liquid crystals.

(a) Smectic (Greek: = Soap- like) liquid crystals:

(1) These have molecules parallel to each other arranged in layers. The smectic state is exhibited by common soaps

at higher temperature ~ 200 oC in the presence of water.

(2) The smectic liquid crystals do not flow like normal liquids. They have limited mobility.

(3) They flow in layers and the distribution of velocity is different in different layers i.e. the flow of smectic liquid crystals

is non-Newtonian. The viscosity concept is not applicable to these crystals.

(4) The smectic liquid crystals give X-ray diffraction patterns like solid crystals but in one direction only.

e.g. ethyl- p- azoxy benzoate, ethyl- p- azoxy cinnamate etc.

(b) Nematic crystals:

(1) These have molecules parallel to each other and free to slide or roll individually (Greek: Nematic = thread).

(2) They flow like normal liquids and flow readily than smectic liquid crystals. The flow of these liquid crystals obeys the Newtonian laws. Their viscosity is low as compared to liquids.

(3) These are closer to true anisotropic liquids than smectic liquid crystals. They have low viscosity.

(4) In polarized light they have mobile thread like structure. The molecules are aggregated together in groups or swarms with their axis parallel to one another.

(5) In electric or magnetic field, the swarms of nematic liquid crystals are oriented in unique direction.

(6) The turbidity of nematic crystals arises due to the scattering of light by these crystals.

(7) Due to orientation of molecules about a preferred axis (director axis) nematic state shows anisotropy in optical, diamagnetic, elastic properties etc. e.g. p- azoxy phenetole, p-azoxy anisole, dibenzal benzidine etc.

(c) Chloesteric liquid crystals:

In these crystals, the molecules are arranged parallel in the form of layers. The molecules in successive layers are slightly rotated with the layers above and below forming a spiral structure. They exhibit very high optical rotatory power. The pitch of spiral and reflected color depends on temperature.

They are named as chloesteric crystals because the skeleton of substance passing through this state is as like that of cholesterol.

Chloesteric liquid crystals have some smectic and some nematic characters. e.g. cholesteryl benzoate (m.p. = 178 oC, transition point = 145 oC).

Application of Liquid Crystals

1. Liquid crystal displays are common in calculators, digital watches, television displays using L.C. screens has also been developed. Cholesteric liquid crystals have also been used for novelty items such as toys and decorative materials.

2.liquid crystals are widely used in cosmetic industry in manufacturing of liquid crystal makeup removers, lipsticks and lip glasses containing cholesteric liquid crystals.

3. Liquid crystals are using extensively in pharmaceutical industries.

4.Helmets and bullet proof jackets

5.Ploymer industry

6. Cholesteric liquid crystal substances, when applied to the surface of the skin, have been used to locate veins, arteries, infections, tumors and the fetal placenta which are warmer than the surrounding tissues.

7. Cosmetics: Used in makeup removers, lipsticks, and other cosmetic products for their specific textures and iridescent properties.

Thank you